

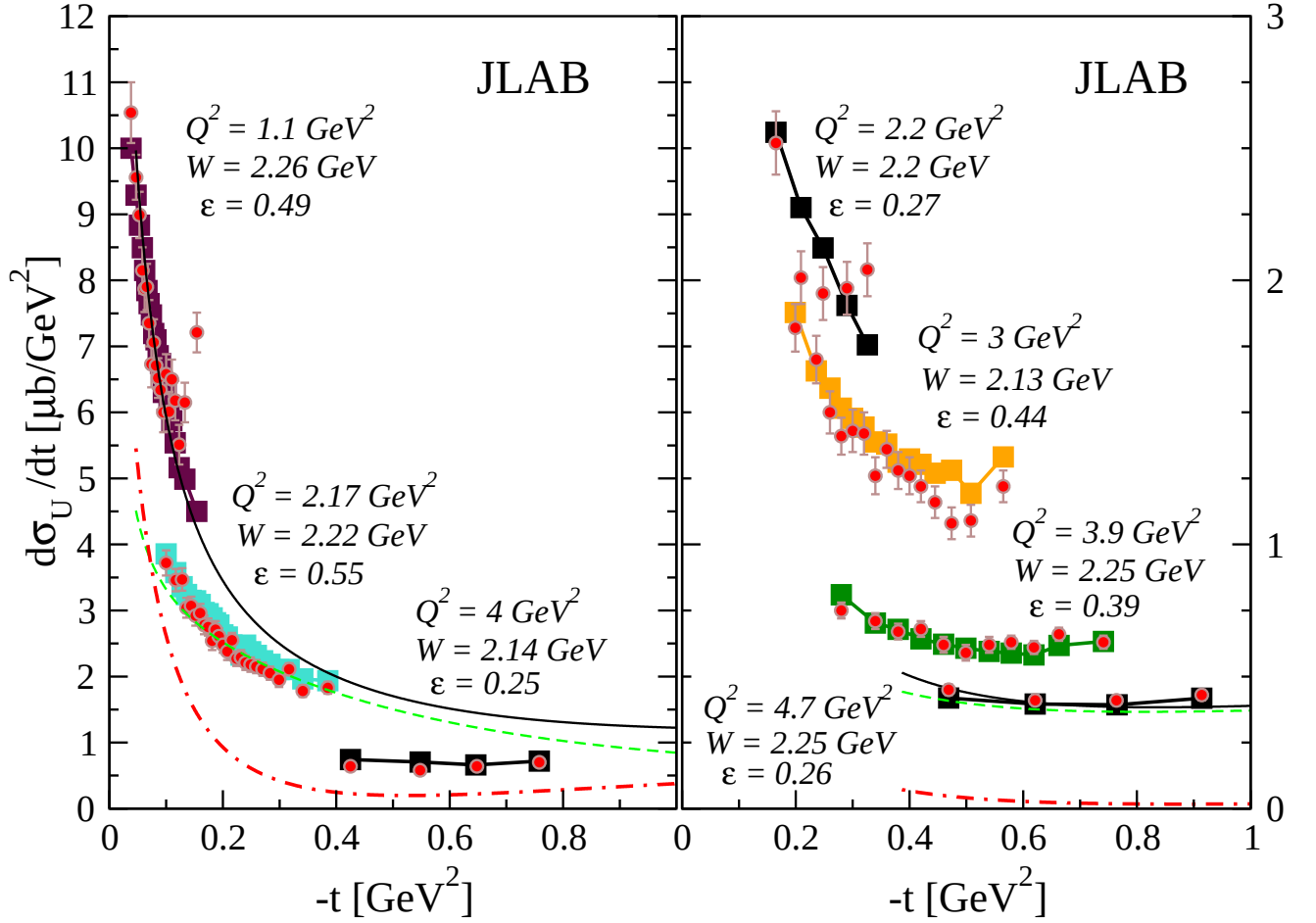


FIG. 9: (Color online) The differential cross sections $d\sigma_U/dt = d\sigma_T/dt + \varepsilon d\sigma_L/dt$ in exclusive reaction $p(\gamma^*, \pi^+)n$ in the kinematics of the π -CT experiment at JLAB [52]. The square symbols connected by solid lines describe the model results. The discontinuities in the curves result from the different values of (Q^2, W, ε) for the various $-t$ bins. The dash-dotted and dashed curves describe the contributions of the longitudinal $\varepsilon d\sigma_L$ and transverse $d\sigma_T$ cross sections, respectively, to the total unseparated cross sections (solid curves) for the lowest and highest average values of $Q^2 = 1.1 \text{ GeV}^2$ and $Q^2 = 4.7 \text{ GeV}^2$.

formed for values of (Q^2, W) corresponding to the first $-t$ bin [2, 4]. The notations for the curves are the same as in Figure 6. In these calculations we used the value of $\Lambda_{\gamma\pi\pi}^2 = 0.4 \text{ GeV}^2$. Also here we find a pronounced resonance contribution in $d\sigma_T/dt$, $d\sigma_{TT}/dt$ and $d\sigma_{LT}/dt$. The slope and magnitude of $d\sigma_L/dt$ at forward angles are also affected by the resonances.

An extraction of L/T partial differential cross sections requires besides the Rosenbluth separation a fit of different harmonics in the azimuthal ϕ -angle distribution of the measured unseparated double differential cross sections. In the actual experiment one measures $d^2\sigma/dtd\phi$ for two different ε bins. In Figure 8 we show the ϕ dependence of $2\pi d^2\sigma/dtd\phi$ in the reaction $p(\gamma^*, \pi^+)n$ at fixed $-t$ and two (high and low) values of ε . This is a representative example of ϕ -dependent exclusive cross sections. In Figure 8 the solid curves are the model results and experimental data are from [1–3]. As in Figures 6 the dashed curves correspond to the contribution

of the π -exchange and dash-dotted curves do not account for the resonances. The upper set of solid, dashed and dash-dotted curves belongs to the higher value of ε .

In Figure 9 we confront the result of our calculations with the new JLAB $p(\gamma^*, \pi^+)n$ data [52] for unseparated cross sections $d\sigma_U/dt$, see Eq. (6), at values of $W \simeq 2.2 \div 2.4 \text{ GeV}$ and for different values of (Q^2, ε) bins. The square symbols connected by solid lines describe the model results. The discontinuities in the curves result from the different values of (Q^2, W, ε) for the various $-t$ bins. The data are very well reproduced by the present model in the measured Q^2 range from $Q^2 \simeq 1 \text{ GeV}^2$ up to 5 GeV^2 . In Figure 9 we also show the contributions of the longitudinal $\varepsilon d\sigma_L$ (dash-dotted curves) and transverse $d\sigma_T$ (dashed curves) cross sections to the total unseparated cross sections (solid curves) for the lowest and highest average values of $Q^2 = 1.1 \text{ GeV}^2$ and $Q^2 = 4.7 \text{ GeV}^2$. The cross sections at high values of Q^2 are flat and totally transverse. At forward angles a strong